Project Design Phase Problem – Solution Fit

Date	11/07/26
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
Maximum Marks	2 Marks

Problem – Solution :

Problem:

College students often adopt poor dietary habits due to lack of awareness, accessibility to nutritious food, and minimal real-time feedback about their consumption patterns. This can result in nutritional deficiencies, reduced academic performance, and long-term health complications. Institutions lack a centralized, data-driven mechanism to monitor, analyze, and respond to these dietary trends in real time.

Solution:

This project proposes a Tableau-powered platform that visualizes dietary data, exercise patterns, and health perceptions among students. Through real-time data collection, transformation, and interactive visual analytics, stakeholders can proactively identify health risks, monitor student nutrition trends, and implement targeted interventions. Predictive models enable personalized dietary planning, fostering better student health and academic success through data-driven strategies.

Purpose:

- Understand the current dietary behavior and health challenges among college students.
- Deliver personalized, real-time nutritional insights and trends using Tableau dashboards.
- Promote evidence-based decisions for meal planning, health services, and awareness campaigns.
- Enable educational institutions to respond proactively to deficiencies or risky trends.
- Support students with tailored dietary suggestions to enhance overall well-being and academic outcomes.

Template:


Comprehensive Analysis and Dietary Strategies with Tableau
A College Food Choices Case Study - Revolutionizing dietary data visualization to drive informed decision-making and enhance student health and academic performance
1 KEY STAKEHOLDERS
KS
University administration and health services
Student nutrition counselors and dietitians
Cafeteria and food service management
Student wellness programs coordinators
Academic performance tracking departments
2 CURRENT CHALLENGES
CC
Limited visibility into student dietary patterns
Difficulty tracking nutritional intake trends
Lack of real-time dietary monitoring systems
Insufficient data for evidence-based nutrition programs
Poor correlation analysis between diet and academic performance
3 PROJECT TRIGGERS
PT
Rising concerns about student health and nutrition
Need for data-driven dietary intervention strategies
Growing awareness of diet-academic performance correlation
University mandate for comprehensive student wellness tracking
Availability of advanced visualization tools like Tableau
4 USE CASE SCENARIOS
UCS
Real-time monitoring of nutritional intake patterns
Swift response to widespread dietary deficiencies
Predictive analysis for personalized nutrition plans
Proactive identification of at-risk student populations
Evidence-based cafeteria menu optimization
5 TABLEAU SOLUTIONS
TS
Interactive dashboards for dietary trend visualization
Real-time alerts for nutritional intake anomalies
Predictive analytics for health risk assessment
Comprehensive student dietary profile creation
Automated reporting for health service interventions
6 EXPECTED BENEFITS
EB
Enhanced student health and well-being outcomes
Improved academic performance through better nutrition
Reduced healthcare costs through preventive measures
Data-driven decision making for university policies
Increased student satisfaction with campus dining
7 KEY DATA INSIGHTS
KDI
Dietary pattern correlation with academic performance
Nutritional deficiency prevalence across demographics
Seasonal variations in food consumption habits
Exercise habits impact on dietary choices
Health perception influence on eating behaviors
8 TARGET OUTCOMES
TO
20% improvement in student nutritional awareness
15% increase in healthy food choices on campus
Reduction in diet-related health issues by 25%
Enhanced cafeteria menu satisfaction scores
Establishment of evidence-based nutrition programs
9 IMPLEMENTATION STRATEGY
IS
Phase 1: Data collection and Tableau dashboard setup
Phase 2: Pilot testing with select student groups
Phase 3: Full deployment across campus
Phase 4: Training for staff and stakeholders
Phase 5: Continuous monitoring and optimization